

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Product Description: | <u>Cyclohexane</u>                                    |
| Product Grade:       | ER, SQ, HPLC                                          |
| Cat No. :            | Q18245, Q18247, Q22855, Q22857, Q2285C, Q43406        |
| Synonyms             | Hexahydrobenzene; Benzene hexahydride; Hexamethylene. |
| Index No             | 601-017-00-1                                          |
| CAS No               | 110-82-7                                              |
| EC No                | 203-806-2                                             |
| Molecular Formula    | C <sub>6</sub> H <sub>12</sub>                        |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                      |                          |
|----------------------|--------------------------|
| Recommended Use      | Laboratory chemicals.    |
| Uses advised against | No Information available |

### 1.3. Details of the supplier of the safety data sheet

|         |                                                                                                                                   |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|
| Company | Thermo Fisher Scientific India Pvt. Ltd<br>403-404, B-wing, Delphi,<br>Hiranandani Business Park,<br>Powai, Mumbai 4000076, INDIA |
|---------|-----------------------------------------------------------------------------------------------------------------------------------|

|                |                                     |
|----------------|-------------------------------------|
| E-mail address | laboratorysolution@thermofisher.com |
|----------------|-------------------------------------|

### 1.4. Emergency telephone number

India Toll Free: 18 00 22 22 30  
Chemtrec US: (800) 424-9300  
Chemtrec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

#### CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567

##### Physical hazards

|                   |                   |
|-------------------|-------------------|
| Flammable liquids | Category 2 (H225) |
|-------------------|-------------------|

##### Health hazards

|                           |                   |
|---------------------------|-------------------|
| Aspiration Toxicity       | Category 1 (H304) |
| Skin Corrosion/Irritation | Category 2 (H315) |

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Specific target organ toxicity - (single exposure)

Category 3 (H336)

## Environmental hazards

Acute aquatic toxicity  
Chronic aquatic toxicity

Category 1 (H400)  
Category 1 (H410)

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Danger

## Hazard Statements

H225 - Highly flammable liquid and vapor  
H304 - May be fatal if swallowed and enters airways  
H315 - Causes skin irritation  
H336 - May cause drowsiness or dizziness  
H410 - Very toxic to aquatic life with long lasting effects

## Precautionary Statements

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P301 + P310 - IF SWALLOWED: Immediately call a POISON CENTER or doctor/physician  
P331 - Do NOT induce vomiting  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)

Toxic to terrestrial vertebrates

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component   | CAS No   | EC No     | Weight % | CLP Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567                         |
|-------------|----------|-----------|----------|-----------------------------------------------------------------------------------------------------------------|
| Cyclohexane | 110-82-7 | 203-806-2 | >95      | Flam. Liq. 2 (H225)<br>Asp. Tox. 1 (H304)<br>Skin Irrit. 2 (H315)<br>STOT SE 3 (H336)<br>Aquatic Acute 1 (H400) |

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|  |  |  |  |                          |
|--|--|--|--|--------------------------|
|  |  |  |  | Aquatic Chronic 1 (H410) |
|--|--|--|--|--------------------------|

| Component   | Specific concentration limits (SCL's) | M-Factor | Component notes |
|-------------|---------------------------------------|----------|-----------------|
| Cyclohexane | -                                     | 1        | -               |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

### 4.1. Description of first aid measures

|                                           |                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                           |
| <b>Skin Contact</b>                       | Wash off immediately with soap and plenty of water for at least 15 minutes. Get medical attention.                                                                                                                                                                                                                                                                                        |
| <b>Ingestion</b>                          | Do NOT induce vomiting. Aspiration hazard. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                         |
| <b>Inhalation</b>                         | Remove to fresh air. If breathing is difficult, give oxygen. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Aspiration into lungs can produce severe lung damage. Get medical attention immediately if symptoms occur. |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                          |

### 4.2. Most important symptoms and effects, both acute and delayed

Difficulty in breathing. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

### 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

#### Extinguishing media which must not be used for safety reasons

Do not use a solid water stream as it may scatter and spread fire.

### 5.2. Special hazards arising from the substance or mixture

Flammable. Risk of ignition. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Containers may explode when heated. Do not allow run-off from fire-fighting to enter drains or water courses.

#### Hazardous Combustion Products

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Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

## **5.3. Advice for firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## **SECTION 6: ACCIDENTAL RELEASE MEASURES**

### **6.1. Personal precautions, protective equipment and emergency procedures**

Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### **6.2. Environmental precautions**

Avoid release to the environment. Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

### **6.3. Methods and material for containment and cleaning up**

Remove all sources of ignition. Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

### **6.4. Reference to other sections**

Refer to protective measures listed in Sections 8 and 13.

## **SECTION 7: HANDLING AND STORAGE**

### **7.1. Precautions for safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not breathe mist/vapors/spray. Avoid contact with skin, eyes or clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Use spark-proof tools and explosion-proof equipment. Take precautionary measures against static discharges.

#### **Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### **7.2. Conditions for safe storage, including any incompatibilities**

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Flammables area.

**Technical Rules for Hazardous Substances (TRGS) 510**      Class 3  
**Storage Class (LGK) (Germany)**

### **7.3. Specific end use(s)**

Use in laboratories

## **SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION**

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## 8.1. Control parameters

### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component   | The United Kingdom                                                                                                  | European Union                                         | Ireland                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Cyclohexane | STEL: 300 ppm 15 min<br>STEL: 1050 mg/m <sup>3</sup> 15 min<br>TWA: 100 ppm 8 hr<br>TWA: 350 mg/m <sup>3</sup> 8 hr | TWA: 200 ppm (8hr)<br>TWA: 700 mg/m <sup>3</sup> (8hr) | TWA: 200 ppm 8 hr.<br>TWA: 700 mg/m <sup>3</sup> 8 hr.<br>STEL: 600 ppm 15 min<br>STEL: 2100 mg/m <sup>3</sup> 15 min |

### Biological limit values

List source(s):

### Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                       | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|---------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Cyclohexane<br>110-82-7 ( >95 ) |                              |                                 |                                | DNEL = 2016mg/kg bw/day           |

| Component                       | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation) | Chronic effects systemic (Inhalation) |
|---------------------------------|----------------------------------|-------------------------------------|------------------------------------|---------------------------------------|
| Cyclohexane<br>110-82-7 ( >95 ) | DNEL = 1400mg/m <sup>3</sup>     | DNEL = 1400mg/m <sup>3</sup>        | DNEL = 700mg/m <sup>3</sup>        | DNEL = 700mg/m <sup>3</sup>           |

### Predicted No Effect Concentration (PNEC)

See values below.

| Component                       | Fresh water      | Fresh water sediment          | Water Intermittent | Microorganisms in sewage treatment | Soil (Agriculture)       |
|---------------------------------|------------------|-------------------------------|--------------------|------------------------------------|--------------------------|
| Cyclohexane<br>110-82-7 ( >95 ) | PNEC = 0.207mg/L | PNEC = 16.68mg/kg sediment dw | PNEC = 0.207mg/L   | PNEC = 3.24mg/L                    | PNEC = 3.38mg/kg soil dw |

| Component                       | Marine water     | Marine water sediment         | Marine water intermittent | Food chain | Air |
|---------------------------------|------------------|-------------------------------|---------------------------|------------|-----|
| Cyclohexane<br>110-82-7 ( >95 ) | PNEC = 0.207mg/L | PNEC = 16.68mg/kg sediment dw |                           |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

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| Hand Protection          |                   | Protective gloves                                                         |             |                                                                                |
|--------------------------|-------------------|---------------------------------------------------------------------------|-------------|--------------------------------------------------------------------------------|
| Glove material           | Breakthrough time | Glove thickness                                                           | EU standard | Glove comments                                                                 |
| Nitrile rubber           | > 480 minutes     | 0.38 - 0.56 mm                                                            | Level 6     | As tested under EN374-3 Determination of Resistance to Permeation by Chemicals |
| Viton (R)                | > 480 minutes     | 0.7 mm                                                                    | EN 374      |                                                                                |
| Neoprene gloves          | < 240 minutes     | 0.45 mm                                                                   |             |                                                                                |
| Skin and body protection |                   | Wear appropriate protective gloves and clothing to prevent skin exposure. |             |                                                                                |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

|                                        |                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Respiratory Protection</b>          | When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.<br>To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly                                                                                     |
| <b>Large scale/emergency use</b>       | Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced<br><b>Recommended Filter type:</b> Organic gases and vapours filter Type A Brown conforming to EN14387                                                                  |
| <b>Small scale/Laboratory use</b>      | Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.<br><b>Recommended half mask:-</b> Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141<br>When RPE is used a face piece Fit Test should be conducted |
| <b>Environmental exposure controls</b> | Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.                                                                                                                                           |

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                                |                                                |                                 |
|------------------------------------------------|------------------------------------------------|---------------------------------|
| <b>Physical State</b>                          | Liquid                                         |                                 |
| <b>Appearance</b>                              | Colorless                                      |                                 |
| <b>Odor</b>                                    | sweet                                          |                                 |
| <b>Odor Threshold</b>                          | No data available                              |                                 |
| <b>Melting Point/Range</b>                     | 6.5 °C / 43.7 °F                               |                                 |
| <b>Softening Point</b>                         | No data available                              |                                 |
| <b>Boiling Point/Range</b>                     | 81 °C / 177.8 °F                               |                                 |
| <b>Flammability (liquid)</b>                   | Highly flammable                               | On basis of test data           |
| <b>Flammability (solid,gas)</b>                | Not applicable                                 | Liquid                          |
| <b>Explosion Limits</b>                        | <b>Lower</b> 1.2 vol%<br><b>Upper</b> 8.4 vol% |                                 |
| <b>Flash Point</b>                             | -18 °C / -0.4 °F                               | <b>Method -</b> CC (closed cup) |
| <b>Autoignition Temperature</b>                | 260 °C / 500 °F                                |                                 |
| <b>Decomposition Temperature</b>               | No data available                              |                                 |
| <b>pH</b>                                      | No information available                       |                                 |
| <b>Viscosity</b>                               | 0.94 mPa.s @ 20 °C                             |                                 |
| <b>Water Solubility</b>                        | practically insoluble                          | 0.052 g/l                       |
| <b>Solubility in other solvents</b>            | No information available                       |                                 |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                |                                 |
| <b>Component</b>                               | log Pow                                        |                                 |

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|                            |                         |             |
|----------------------------|-------------------------|-------------|
| Cyclohexane                | 3.44                    |             |
| Vapor Pressure             | 104 mbar @ 20 °C        |             |
| Density / Specific Gravity | 0.770                   |             |
| Bulk Density               | Not applicable          | Liquid      |
| Vapor Density              | 2.90                    | (Air = 1.0) |
| Particle characteristics   | Not applicable (liquid) |             |

## 9.2. Other information

|                      |                                             |
|----------------------|---------------------------------------------|
| Molecular Formula    | C6 H12                                      |
| Molecular Weight     | 84.15                                       |
| Explosive Properties | Vapors may form explosive mixtures with air |
| Evaporation Rate     | 6.1 - (Butyl Acetate = 1.0)                 |

## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Stable under normal conditions.

### 10.3. Possibility of hazardous reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Keep away from open flames, hot surfaces and sources of ignition.

### 10.5. Incompatible materials

Strong oxidizing agents.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

|            |                                                                  |
|------------|------------------------------------------------------------------|
| Oral       | Based on available data, the classification criteria are not met |
| Dermal     | Based on available data, the classification criteria are not met |
| Inhalation | Based on available data, the classification criteria are not met |

| Component   | LD50 Oral            | LD50 Dermal             | LC50 Inhalation                            |
|-------------|----------------------|-------------------------|--------------------------------------------|
| Cyclohexane | > 5000 mg/kg ( Rat ) | > 2000 mg/kg ( Rabbit ) | LC50 > 32880 mg/m <sup>3</sup> ( Rat ) 4 h |

#### (b) skin corrosion/irritation;

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(c) serious eye damage/irritation; Based on available data, the classification criteria are not met

(d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

(e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

(f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity;

Based on available data, the classification criteria are not met

(h) STOT-single exposure;

Category 3

Results / Target organs

Central nervous system (CNS).

(i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

(j) aspiration hazard;

Category 1

Symptoms / effects, both acute and delayed

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## 11.2. Information on other hazards

Endocrine Disrupting Properties

Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component   | Freshwater Fish                                                                                                                                                                                                                                          | Water Flea          | Freshwater Algae   |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|
| Cyclohexane | LC50: 48.87 - 68.76 mg/L, 96h static (Poecilia reticulata)<br>LC50: 24.99 - 44.69 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 23.03 - 42.07 mg/L, 96h static (Pimephales promelas)<br>LC50: 3.96 - 5.18 mg/L, 96h flow-through (Pimephales promelas) | EC50 = 0.9 mg/L/48h | EC50 >500 mg/L/72h |

| Component   | Microtox                                        | M-Factor |
|-------------|-------------------------------------------------|----------|
| Cyclohexane | EC50 = 85.5 mg/L 5 min<br>EC50 = 93 mg/L 10 min | 1        |

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## **12.2. Persistence and degradability**

Readily biodegradable

### **Persistence**

Persistence is unlikely, based on information available.

| Component                       | Degradability |
|---------------------------------|---------------|
| Cyclohexane<br>110-82-7 ( >95 ) | 77% (28d)     |

### **Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## **12.3. Bioaccumulative potential**

Bioaccumulation is unlikely

| Component   | log Pow | Bioconcentration factor (BCF) |
|-------------|---------|-------------------------------|
| Cyclohexane | 3.44    | 83.15                         |

## **12.4. Mobility in soil**

The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

## **12.5. Results of PBT and vPvB assessment**

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

## **12.6. Endocrine disrupting properties**

### **Endocrine Disruptor Information**

This product does not contain any known or suspected endocrine disruptors

## **12.7. Other adverse effects** **Persistent Organic Pollutant** **Ozone Depletion Potential**

This product does not contain any known or suspected substance

This product does not contain any known or suspected substance

## **SECTION 13: DISPOSAL CONSIDERATIONS**

### **13.1. Waste treatment methods**

#### **Waste from Residues/Unused Products**

Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

#### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.

#### **European Waste Catalogue (EWC)**

According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

#### **Other Information**

Do not flush to sewer. Waste codes should be assigned by the user based on the application for which the product was used. Can be landfilled or incinerated, when in compliance with local regulations. Do not let this chemical enter the environment. Do not empty into drains.

## **SECTION 14: TRANSPORT INFORMATION**

### **IMDG/IMO**

#### **14.1. UN number**

UN1145

QUAQ22855

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**14.2. UN proper shipping name** Cyclohexane  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

## ADR

**14.1. UN number** UN1145  
**14.2. UN proper shipping name** Cyclohexane  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

## IATA

**14.1. UN number** UN1145  
**14.2. UN proper shipping name** Cyclohexane  
**14.3. Transport hazard class(es)** 3  
**14.4. Packing group** II

**14.5. Environmental hazards** Dangerous for the environment  
 Product is a marine pollutant according to the criteria set by IMDG/IMO

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

### 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

#### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component   | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Cyclohexane | 110-82-7 | 203-806-2 | -      | -   | X     | X    | KE-18562 | X    | X    |

| Component   | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Cyclohexane | 110-82-7 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

| Component   | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------|----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Cyclohexane | 110-82-7 | -                                                                   | Use restricted. See item 57.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

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## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Seveso III Directive (2012/18/EC)

| Component   | CAS No   | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Major Accident<br>Notification | Seveso III Directive (2012/18/EC) -<br>Qualifying Quantities for Safety Report<br>Requirements |
|-------------|----------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| Cyclohexane | 110-82-7 | Not applicable                                                                                  | Not applicable                                                                                 |

## Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

## Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

## National Regulations

**UK** - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

## WGK Classification

See table for values

| Component   | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-------------|---------------------------------------|-------------------------|
| Cyclohexane | WGK2                                  |                         |

| Component   | France - INRS (Tables of occupational diseases)      |
|-------------|------------------------------------------------------|
| Cyclohexane | Tableaux des maladies professionnelles (TMP) - RG 84 |

| Component                       | Switzerland - Ordinance on the<br>Reduction of Risk from<br>handling of hazardous<br>substances preparation (SR<br>814.81) | Switzerland - Ordinance on<br>Incentive Taxes on Volatile<br>Organic Compounds (OVOC) | Switzerland - Ordinance of the<br>Rotterdam Convention on the<br>Prior Informed Consent<br>Procedure |
|---------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Cyclohexane<br>110-82-7 ( >95 ) | Prohibited and Restricted<br>Substances                                                                                    | Group I                                                                               |                                                                                                      |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H225 - Highly flammable liquid and vapor

H304 - May be fatal if swallowed and enters airways

H315 - Causes skin irritation

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Cyclohexane

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H336 - May cause drowsiness or dizziness  
H400 - Very toxic to aquatic life  
H410 - Very toxic to aquatic life with long lasting effects

## Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

## Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Creation Date** 06-Oct-2009

**Revision Date** 11-Oct-2023

**Revision Summary** Next Revision Date - Oct 2028.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

## Disclaimer

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**End of Safety Data Sheet**